

Stochastic Models → CHEAT SHEET

Properties:

1. **[Fitting first]:** $\frac{\lambda_i}{\sum \lambda_j}$
2. $P(\min\{x_1, \dots, x_n\} > x) = \exp\left[-\sum_{i=1}^n \lambda_i x\right]$
3. $\lambda_1 + \lambda_2 + \lambda_3 + \dots = \lambda_{total}$

$\Gamma_n \sim \text{Gamma}$

Proof (counting process)

$$P(N_t = k) = P(N_t \leq k) - P(N_t > k) \\ = P(S_k \leq t) - P(S_{k+1} \leq t)$$

$$P(S_k \leq t) = \int_0^t \text{Gamma} \cdot dx \\ = \frac{(\lambda t)^k}{k!} e^{-\lambda t} + P(S_{k+1} \leq t)$$

$$\therefore P(S_k < t) = P(S_{k+1} < t) + \frac{(\lambda t)^k}{k!} e^{-\lambda t}$$

inhomogeneous

$$P(N_t = k) = \frac{(\int_0^t \lambda(x) dx)^k}{k!} \exp\left(-\int_0^t \lambda(x) dx\right)$$

$$P(N(0) = 0) = \frac{(\lambda_0)^0}{0!} e^{-\lambda_0} = e^{-\lambda_0}$$

$$P(N(0) = 1) = \lambda_0 (1 - \lambda_0)$$

$$P(N(0) > 2) = 1 - 1 + \lambda_0 - \frac{(\lambda_0)^2}{2}$$

$$P(N(0) > 2) = \frac{(\lambda_0)^2}{2} \cdot \frac{1}{0!}$$

Proof of poisson splitting

$$P(N_t^1 = k_1, N_t^2 = k_2 | N_t^1 + N_t^2 = k_1 + k_2)$$

$$= \frac{\binom{k_1+k_2}{k_1} \frac{(\lambda_1 t)^{k_1}}{k_1!} e^{-\lambda_1 t} \frac{(\lambda_2 t)^{k_2}}{k_2!} e^{-\lambda_2 t}}{\frac{(\lambda_1 + \lambda_2 t)^{k_1+k_2}}{(k_1+k_2)!} e^{-(\lambda_1 + \lambda_2)t}}$$

$$= \frac{(\lambda_1 t)^{k_1}}{k_1!} e^{-\lambda_1 t} \frac{(\lambda_2 t)^{k_2}}{k_2!} e^{-\lambda_2 t}$$

$$= P(N_t^1 = k_1) \cdot P(N_t^2 = k_2)$$

$$= P(N_t^1 = k_1) \cdot P(N_t^2 = k_2)$$

Poisson splitting rates

$$P(N_t^1 = k_1, N_t^2 = k_2) / \text{Joint dist}$$

Compound poisson

$$E[N_t] = E[N_t] \cdot E[\gamma_i] \\ = \lambda t \cdot E[\gamma_i]$$

$$\text{Var}[N_t] = \lambda t \cdot E[\gamma_i^2]$$

ORDER Statistics

$$P(S_1 < R | N_t = 1) = \frac{1}{t}$$

$$P(S_1 < S | N_t > 1)$$

$$P(N_t = 1)$$

$$P(N_t = 1, N_t - N_s = 0)$$

$$P(N_t = 1)$$

$$P(N_t = 1)$$

$$P(N_t = 1)$$

$$P(N_t = 1)$$

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CTMC: $\emptyset \rightarrow 1 \rightarrow 2 \rightarrow 3$

arrival $\rightarrow \lambda$
deft $\rightarrow \mu$

0 arrival 0 depart $[+0]$ $\rightarrow \lambda$
0 arrival 1 depart $[-1]$ $\rightarrow \mu$
1 arrival 0 depart $[+1]$ $\rightarrow \lambda$
1 arrival 1 depart $[+0]$ $\rightarrow \lambda, \mu$

For state 0 $\rightarrow 1$:

$$\pi_0(\lambda + 0) = \pi_1(\mu + 0)$$

$$\pi_0 \lambda = \pi_1 \mu \therefore \pi_1 = \frac{\lambda}{\mu} \pi_0$$

$$\pi_1(\mu + 0) = \pi_2(\lambda + 0)$$

$$\pi_1 \mu = \pi_2 \lambda \therefore \pi_2 = \frac{\lambda}{\mu} \pi_1$$

$$\pi_2 = \frac{\lambda}{\mu} \pi_1 = \left(\frac{\lambda}{\mu}\right)^2 \pi_0$$

$$\text{Again } \sum \pi_n = 1$$

$$\therefore \sum_{n=0}^{\infty} \pi_0 \left(\frac{\lambda}{\mu}\right)^n = \pi_0 \frac{1}{1 - \lambda/\mu}$$

$$\therefore \pi_0 = 1 - \lambda/\mu$$

For a server

Arrival rate $\rightarrow \lambda$

Traffic $\rho = \lambda/\mu$

Case 1: $n < c$

$$\pi_n = n \pi_1 \therefore \pi_{n+1} \cdot \mu_{n+1} = \lambda \pi_n$$

$$\pi_{n+1} = \frac{\lambda}{\mu_{n+1}} \pi_n$$

$$\pi_n = \frac{\lambda^n}{n! \mu^n} \pi_0$$

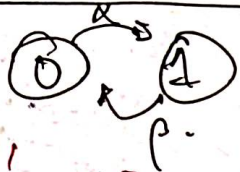
$$\pi_n = \frac{(\lambda/\mu)^n}{n!} \pi_0$$

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Two state GTC



$$P = \begin{bmatrix} \alpha & \beta \\ \alpha & \beta \end{bmatrix}$$

$$P'_{00} = \alpha P_{10} + \alpha P_{00}$$

$$P'_{10} = -\beta P_{10} + \beta P_{00}$$

$$\Delta = P_{10} - P_{00}$$

$$\Delta' = -(\alpha + \beta)P_{00} - (\alpha + \beta)P_{10}$$

$$\Delta' = -(\alpha + \beta)\Delta$$

$$\Delta = \Delta(0) e^{-(\alpha + \beta)t}$$

$$P_{10} = P_{00} + \Delta(0) e^{-(\alpha + \beta)t}$$

$$P_{10} = \beta(P_{00} - P_{10})$$

$$= \beta(P_{00} - P_{00} - \Delta(0) e^{-(\alpha + \beta)t})$$

$$= -\beta \Delta(0) e^{-(\alpha + \beta)t}$$

$$P_{10}(t) = P_{10}(0) + \frac{\beta}{\alpha + \beta} \Delta(0) (1 - e^{-(\alpha + \beta)t})$$

$$P_{00}(t) = P_{00}(0) - \frac{\alpha}{\alpha + \beta} \Delta(0) (1 - e^{-(\alpha + \beta)t})$$

$$E\left[\frac{a + b u_{n+1}}{a + b u_n} \mid x_1, \dots, x_n\right]$$

$$= \frac{a + b u_n + E[x_{n+1} \mid x_1, \dots, x_n]}{a + b u_n + 1}$$

$$= \frac{a + b u_n + 1 \cdot \left(\frac{a + b u_n}{a + b u_n} + 0 \cdot \frac{1}{a + b u_n}\right)}{a + b u_n + 1}$$

$$= \frac{a + b u_n + 1}{a + b u_n + 1} = 1$$

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~~For a Markov chain~~

memoryless holding

$$P(T_i > t + s \mid T_i > s) = P(T_i > t)$$

$$= P(T_i > t)$$

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Markov chain:

$$E[x_{n+1} \mid x_1, \dots, x_n] = x_n$$

sufficient

tower identities:

$$1. E[E[x \mid \mathcal{F}_t] \mid \mathcal{F}_s] = E[x \mid \mathcal{F}_s]$$

$$2. E[E[x \mid \mathcal{F}_t]] = E[x]$$

$$3. E[E[A \mid \mathcal{B}]] = E[A]$$

$$4. E[E[A \mid \mathcal{B}]] = E[A]$$

$$5. E[E[A \mid \mathcal{B}]] = E[A]$$

$$6. E[E[A \mid \mathcal{B}]] = E[A]$$

$$7. E[E[A \mid \mathcal{B}]] = E[A]$$

$$8. E[E[A \mid \mathcal{B}]] = E[A]$$

$$9. E[E[A \mid \mathcal{B}]] = E[A]$$

$$10. E[E[A \mid \mathcal{B}]] = E[A]$$

$$11. E[E[A \mid \mathcal{B}]] = E[A]$$

$$12. E[E[A \mid \mathcal{B}]] = E[A]$$

$$13. E[E[A \mid \mathcal{B}]] = E[A]$$

Abandonment rate

service rate = μ

Arrival rate = λ

Because equal:

$$\lambda P_0 = \mu P_1$$

$$\text{for } n \geq 1$$

$$\lambda P_{n-1} + (\mu + \theta) P_n$$

$$= [\lambda + \mu + (1 - \theta)] P_n$$

Markov chain stopping

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